

Ahmed Mohamed

AI Engineer / Technical Lead

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📍 Melbourne, Australia

30+ PAPERS

12,138 STUDY PARTICIPANTS

>10 TB DATA ANALYSED

3 BIOCONDUCTOR PKGS

I build LLM systems that go into production and stay there: extraction pipelines, multi-provider structured output, cell-level accuracy evaluation, and the AWS and data-governance setup to run them on sensitive data. Data scientist by training (PhD, 30+ papers, statistics and ML on a 30,000-sample study) and a backend engineer by practice (Go services and financial reporting for an institutional Ethereum operator).

Experience

2023 – now

Co-founder & Technical Lead AIWhisperers, Australia

- Built production LLM systems for SMEs, including a human-reviewed platform converting supplier contracts into structured commercial data for a B2B travel company.
- Wrote the company's LLM tooling: multi-provider structured-output library, composable extraction chains, LLM-run caching, cell-level evaluation comparing prompt strategies and models; DSPy prompt optimisation.
- Established the PHI data-governance framework (Privacy Act 1988, ISO 27001-aligned) and secure AWS architecture: private-subnet SageMaker inference, KMS-encrypted RDS/S3, least-privilege IAM.
- Delivered LLM-for-business workshops for companies in retail, health, tourism and mining.

LLM systems Evals Data governance AWS

2022 – now

Senior Software Engineer Bitwise Onchain Solutions (ex-Attestant)

- Backend engineer for institutional Ethereum and multi-chain staking services.
- Own the client reporting and invoicing system with multi-currency support; maintain client-facing APIs and the multi-chain staking dApp.

Go Ethereum Financial systems

2020 – 2025

Bioinformatician, Senior Research Officer WEHI, Melbourne

- Led experimental design, QC, statistics and ML for the ASPREE proteomics programme: 12,138 participants, ~30,000 LC-MS runs, >10 TB — at the time 10× the largest published MS population proteomics study.
- Built the Nextflow processing pipeline on SLURM HPC (DIA-NN, out-of-memory HDF5 matrices) and a BigQuery-backed QC dashboard watching hundreds of runs a day.
- Designed the analysis-first experimental layout that made batch correction and longitudinal modelling work at scale; ran LASSO biomarker models across 13 conditions.
- Built vissE.cloud end-to-end (React/TypeScript, Flask, Redis, Docker): a public gene-set analysis web server across bulk, single-cell and spatial data; first author of the NAR paper.
- Senior author on an ML urine-spectroscopy test for bladder-cancer recurrence (ROC > 0.8).

Proteomics Statistics & ML Nextflow BigQuery

2017 – 2020

Research Officer QIMR Berghofer & UQ Diamantina, Brisbane

- Created lipidr, an R/Bioconductor package for lipidomics analysis (160+ citations), and LipidSuite, its web server for non-programmers; sole developer of RaftProt V2 — tools accessed tens of thousands of times.
- Statistical lead for collaborator studies; taught R package development and Software Carpentry Python; associate supervisor of a PhD student to completion.

Lipidomics Bioconductor Biostatistics

2012 – 2016

PhD · JSPS Visiting Fellow Kyoto University · Boston University

- NetPathMiner, an R/Bioconductor package for network path mining (Bioinformatics); NMRPro, interactive web processing of NMR spectra (Bioinformatics).
- Pan-cancer analysis of somatic mutations in protein complexes.

Skills

LLM engineering

Extraction pipelines, structured output · Cell-level evals, DSPy prompt optimisation · RAG, multi-provider tooling

Data science

Statistics & ML at population scale · Cancer genomics, Bioconductor · lipidr, NetPathMiner, vissE (author)

Software & infra

Python, Go, TypeScript/React · AWS, Kubernetes, Docker, PostgreSQL · Nextflow, SLURM HPC

Web3

Ethereum staking services · Financial reporting & invoicing

Education

PhD, Bioinformatics & Chemical

Genomics 2012–2016

Kyoto University, Japan

MSc, Advanced Pharmaco-science

2010–2012

Osaka University, Japan

BPharm, Pharmaceutical Sciences

2002–2007

Suez Canal University, Egypt

Awards

- ICR Award for Graduate Students, 2015
- Winner, Novartis BioCamp, 2013
- JSPS International Training Program Fellowship, 2013
- Rotary Yoneyama Memorial Scholarship

Publications

30+ papers. NetPathMiner (Bioinformatics), NMRPro (Bioinformatics), lipidr (160+ citations, in Bioconductor since 2019), vissE.cloud (NAR, first author).

Languages

English, Arabic, Japanese

Projects

2026

Tailor This CV — career knowledge base + agentic tailoring, open source

- Harvest your career once; tailor CVs to any job in minutes. This document is its output.

Claude Code typst

2025

Notebrainer Provenance-first agent memory · Claude Code plugin

- Memory layer where every stored fact traces to its source text.

Postgres pgvector

2024

MapMyMath Procedural maths game · web, iOS, Android

- 1,320 catalogued exam questions; built solo in five weeks.

Python React/WebGL